Kepler’s laws

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use orbital shape, equal areas and period comparisons.

SUCCESS CRITERIA - I can...

1. Place the central body at a focus of an ellipse.
2. Compare speed near and far from the central body.
3. Use $T^2=a^3$ only with the specified units and central body.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Launch at 30 m/s from level ground at 25° or 65°. Use g=10 and no drag. Compare ranges and flight times.

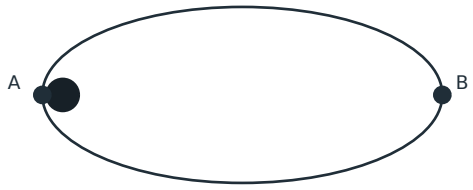
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $T^2 = a^3$.
For Sun orbits only when T is in years and a is in AU. a is the semi-major axis. Equal areas are swept in equal times.

Worked reasoning: For an ideal Sun orbit, a=4 AU gives $T=\sqrt{(4^3)}=8$ years. The orbit period depends on semimajor axis, not on a chosen point's instantaneous speed alone.

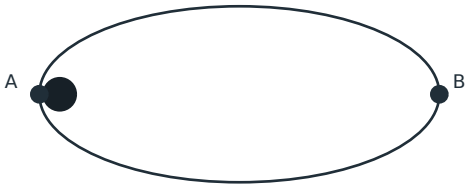
Your turn: Draw an ellipse, put the Sun at a focus and mark the faster and slower parts.



Marzano check - circle: ① ② ③

03 Read and interpret the model

Mark the Sun at a focus on an ellipse. Label where the object is fastest and slowest.



Marzano check - circle: ① ② ③

04 Apply the physics

A Sun-orbit has a=1.42 AU. Compute period. Then find period at a=4 AU and explain why doubling a does not double T.

Semimajor axis (AU)	Period (years)
1.42	____
4	____

Marzano check - circle: ① ② ③

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Continue down each column.

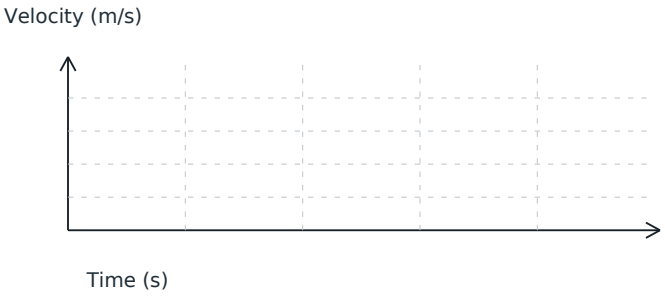
05 Explain your reasoning

Does knowing an orbit’s period alone tell the date when an asteroid reaches Earth’s location? What else is needed?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

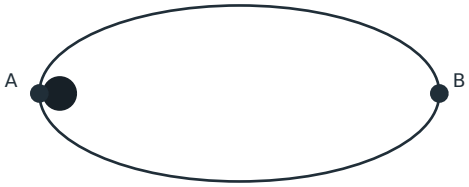
A ball begins with downward velocity 4 m/s and falls 3 s. Use downward positive, g=10 m/s² and no drag. Find final velocity and displacement.
Sketch the motion and label a positive direction.



Marzano check - circle: 1 2 3

07 Exit check

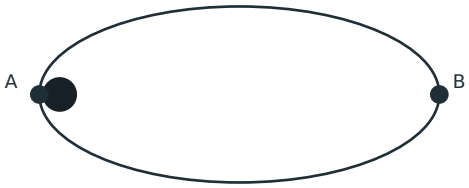
For two Sun-orbits, $a_B=2\,a_A$. Find T_B/T_A. State why the same unit form cannot automatically be used for a Moon orbit about Earth.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

An ideal Sun orbit has period 27 years. Find its semimajor axis. If eccentricity is 0.5, find perihelion and aphelion distances using $r=a(1\mp e)$.



Marzano check - circle: 1 2 3